

EXPERIMENT 1

Design a UI where users recall visual elements (e.g., icons or text chunks). Evaluate the effect of chunking on user memory.

REG NO: 240701324

Name: Mohamed Ismail Fawaz K P

AIM

The aim of this experiment is to evaluate the effect of visual chunking (grouping elements) on a user's short-term memory recall. It specifically tests whether users can recall more items when they are presented in organized groups compared to a random or continuous sequence.

FRAME 1: INSTRUCTION PAGE

- The Instruction screen introduces the Visual Memory Game to the user.
- It clearly explains the objective of the game, which is to test short-term visual memory.
- Users are informed that a set of images will be displayed for a limited time (5 seconds).
- The user must carefully observe and memorize the images within the given time frame.
- After the timer ends, the images will be shuffled and mixed with additional new images.
- The user is required to identify and select only the images that were shown previously.
- Once the selection is completed, the user must click the "Submit" button.
- The system will then display the user's score based on correct selections.
- This screen ensures that users understand the rules before beginning the game.
- It reduces confusion and prepares the user for the interaction flow of the application.

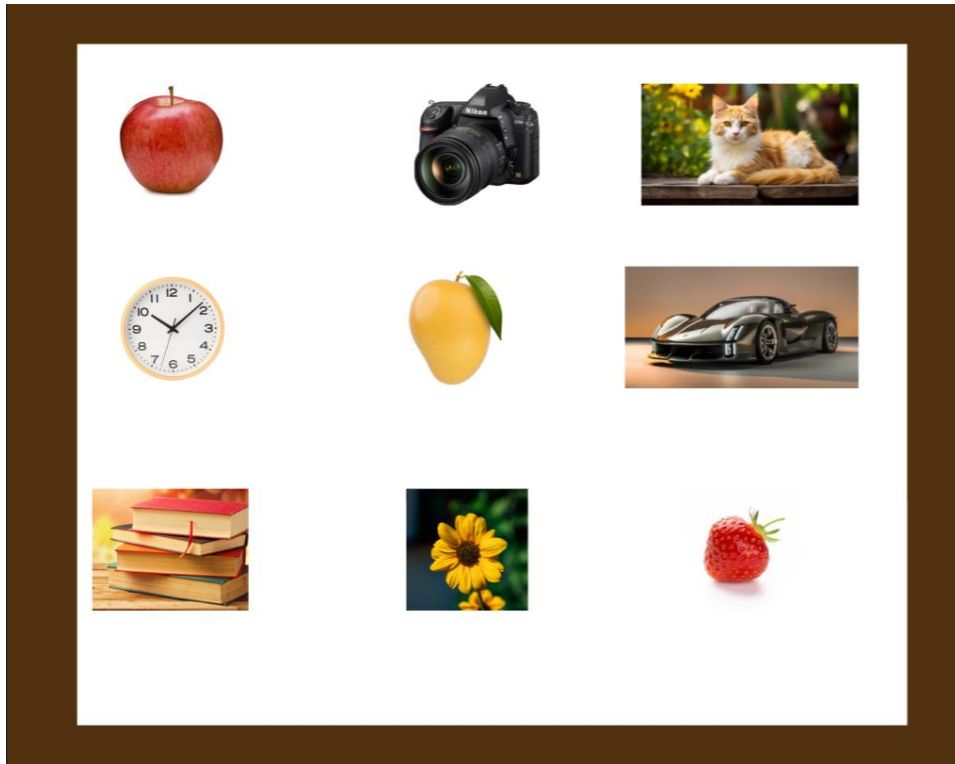
Visual Memory Game

**You will see a set of images for 5
seconds.
Try to remember them carefully.
After that, images will shuffle.
Select the images you saw earlier.**

START

FRAME 2: CHUNKING FACE

- The Memorize screen displays a set of selected images to the user.
- These images are shown for a fixed duration of 5 seconds.
- A visible timer is provided to indicate the remaining time.
- The purpose of this screen is to test the user's short-term visual memory.
- Users are required to carefully observe and remember the images.
- No interaction (such as clicking or selecting) is allowed during this phase.
- The layout is kept clean and distraction-free to help users focus.
- Once the timer reaches zero, the screen automatically transitions to the selection screen.
- This automatic transition ensures fairness and equal time for all users.
- The screen plays a crucial role in evaluating memory retention ability.



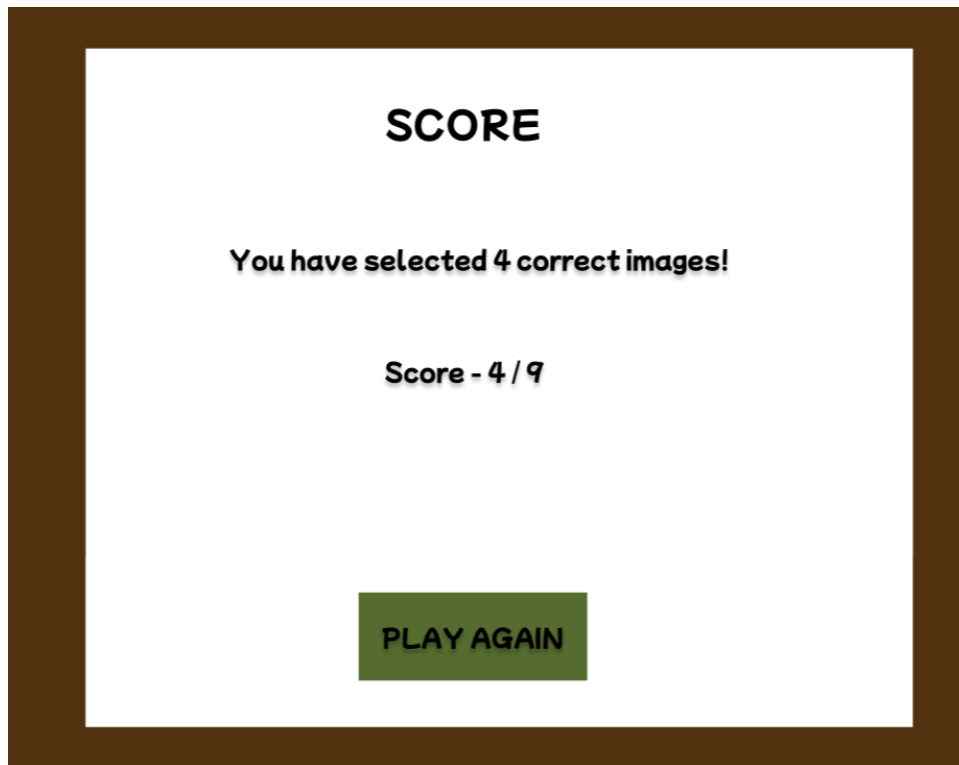
FRAME 3: RECALL PHASE

- The Select/Recall screen appears after the memorization timer ends.
- In this phase, the previously shown images are shuffled.
- Additional new images are added to increase difficulty.
- The arrangement of images is randomized to test actual recall ability.
- Users must carefully identify and select only the images they remember from the previous screen.
- Each image is clickable and changes visually when selected (e.g., highlighted with a border).
- Users can toggle their selection by clicking an image again.
- A “Submit” button is provided to confirm the final selection.
- No time limit is enforced during this phase, allowing users to think carefully.
- This screen evaluates the user’s ability to recall and differentiate between familiar and unfamiliar images.
- The interaction design ensures a smooth and intuitive user experience.
- Upon clicking “Submit,” the system navigates to the Score screen.



FRAME 4: RESULT PAGE

- The Score screen displays the user's performance after submitting their selections.
- It shows the total number of correctly selected images.
- The score is presented in a clear format (e.g., 6/8).
- The screen may also display the accuracy percentage for better understanding.
- A feedback message (e.g., "Great Memory!" or "Try Again!") is shown based on performance.
- The layout is simple and focused to highlight the result.
- This screen provides immediate feedback to the user.
- It helps users understand their memory performance.
- A "Play Again" button is included to restart the game.
- The navigation ensures a smooth return to the Instruction screen.
- The Score screen completes the interaction cycle of the application.
- It reinforces user engagement by encouraging repeated attempts.



Result:

The Visual Memory Game was successfully prototyped with interactive image selection and score display. The application effectively demonstrates short-term visual memory testing.